PlantWhisper: Translating Plant Stress into Sound via Diffusion-Based Ultrasonic Synthesis

A Multi-Modal AI System for Visual Stress Assessment

and Acoustic Sonification of Plant Xylem Cavitation

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Abstract

Plants emit ultrasonic acoustic emissions (UAE) in the 20–150 kHz range during xylem cavitation events induced by drought and other stressors, as demonstrated by Khait et al. (2023). PlantWhisper is a full-stack AI system that bridges this biological phenomenon with computer vision: given a single photograph of a plant, it estimates stress severity, predicts ultrasonic pop rates, and *synthesizes* what those pops would sound like—frequency-shifted into the audible range—using a conditional denoising diffusion probabilistic model (DDPM). The system features (i) FastSAM/SAM segmentation with green-threshold fallback, (ii) MobileNetV2-based disease classification, (iii) Grad-CAM attention visualization, (iv) a stress-conditioned UNet diffusion model generating mel spectrograms of cavitation pops, (v) real-time waveform and spectrogram visualizations, and (vi) LLM-driven empathetic plant speech with TTS rendering. We describe the full architecture, the biological grounding of our stress-to-pop mapping, failure cases encountered during development, and how we handled the absence of a paired image–audio dataset through principled heuristic design. All components run as a containerized FastAPI backend on HuggingFace Spaces with a Next.js portfolio frontend.

1 Introduction

In 2023, Khait et al. published a landmark study in *Cell* demonstrating that plants under drought or physical damage emit airborne sounds in the ultrasonic range (20–250 kHz), primarily caused by cavitation—the formation and collapse of air bubbles in xylem vessels under negative hydraulic pressure [1]. Stressed tomato plants were recorded emitting up to 35 clicks per hour at peak stress, versus roughly 1 click per hour when well-watered.

PlantWhisper asks: *Can we infer what a plant sounds like from a single photograph?* We combine plant disease classification with biologically-grounded acoustic modeling to create an end-to-end system that:

1. Segments the plant from its background
2. Classifies its health condition
3. Estimates a continuous stress level $\sigma \in [0, 1]$
4. Maps stress to an expected pop rate (pops/hour)
5. Synthesizes an audio clip of frequency-shifted cavitation pops
6. Generates waveform and spectrogram visualizations
7. Gives the plant an empathetic "voice" via LLM + TTS

[INSERT FIGURE: System Architecture Diagram]

Full pipeline from image upload through segmentation, classification, stress estimation, audio synthesis, and visualization.

Figure 1: PlantWhisper end-to-end system architecture.

2 Biological Foundation

2.1 Xylem Cavitation and Acoustic Emissions

Water transport in vascular plants occurs through xylem vessels under negative pressure (tension). The cohesion-tension theory [2] describes how transpiration at leaf surfaces pulls water columns upward from the roots through continuous hydrogen bonding. Under drought stress, this tension increases beyond the vessel's structural limits, causing **cavitation**: dissolved gas nucleates into bubbles that expand and collapse violently within the vessel lumen.

Each cavitation event produces a broadband acoustic pulse—an ultrasonic "pop"—with peak energy in the 20–150 kHz range, well above human hearing (∼20 kHz). These are not metabolic signals but *mechanical failure events*: the sound of a plant's vascular system breaking under hydraulic stress.

2.2 What Makes This Biologically Unique

Unlike most plant monitoring approaches (chlorophyll fluorescence, thermal imaging, NDVI), acoustic emissions provide:

- **Real-time stress indication**: Cavitation events occur within minutes of drought onset, far faster than visible wilting (hours to days)
- **Quantitative severity**: Pop rate correlates directly with hydraulic dysfunction—more pops indicate more embolized vessels
- **Species-universal mechanism**: All vascular plants (angiosperms and gymnosperms) share xylem-based water transport; cavitation is a physical inevitability under sufficient tension
- **Pre-visual detection**: A plant may be emitting 20+ clicks/hour while still appearing visually healthy, as initial cavitation affects redundant vessels before causing visible wilting

2.3 The Khait et al. Findings

Key quantitative results from [1] that ground our model:

Table 1: Acoustic emission rates from Khait et al. (2023).

Condition	Pops/Hour	Frequency Range
Well-watered (control)	∼1	20–100 kHz
Drought stress (peak)	∼35	20–150 kHz
Cut stem	∼25	20–100 kHz

We use these values directly as our model's anchors: $R_{\text{healthy}} = 1.0$ pops/hour and $R_{\text{peak}} = 35.0$ pops/hour (see Section 4).

[INSERT FIGURE: Xylem Cavitation Illustration]

Diagram showing water column under tension in xylem vessel, bubble nucleation, and collapse producing an acoustic pulse.

Figure 2: Mechanism of xylem cavitation producing ultrasonic emissions.

2.4 Frequency Shifting: Making the Inaudible Audible

Real cavitation pops peak at ∼53 kHz. Since human hearing cuts off at ∼20 kHz, we apply a conceptual frequency shift:

$$f_{\text{audible}} = f_{\text{real}} \times \frac{f_{\text{target}}}{f_{\text{peak}}} = 53{,}000 \times \frac{1{,}000}{53{,}000} = 1{,}000 \text{ Hz} \tag{1}$$

This maps the peak ultrasonic energy to 1 kHz in our synthesized audio, preserving temporal structure while making the pops clearly audible through standard speakers.

3 System Architecture

PlantWhisper consists of three deployment layers:

1. **Backend API**: FastAPI (Python 3.11) containerized via Docker, deployed on HuggingFace Spaces. Hosts all ML models and audio synthesis.
2. **Frontend**: Next.js 14 (React/TypeScript) portfolio site with glassmorphic UI, deployed separately.
3. **Model stack**: PyTorch models running on CPU (no GPU required at inference).

3.1 Segmentation Pipeline

The system implements a three-tier segmentation cascade for robustness:

Tier 1 — FastSAM (Primary). We use FastSAM-s [3], a YOLO-based architecture that achieves SAM-quality segmentation in real time. The model runs with confidence threshold 0.4 and IoU threshold 0.9. When multiple masks are returned, we select the mask whose centroid is closest to the image center:

$$m^* = \arg\min_{m_i} \left[(\bar{y}_i - h/2)^2 + (\bar{x}_i - w/2)^2\right] \quad (2)$$

where $(\bar{y}_i, \bar{x}_i)$ is the centroid of mask m_i.

Tier 2 — SAM (Fallback). If FastSAM is unavailable, we fall back to Meta's Segment Anything Model (SAM) with ViT-B backbone [4]. A single center-point prompt is used with multi-mask output, selecting the highest-confidence mask.

Tier 3 — Green Threshold (Emergency). If both neural segmentation models fail, we apply HSV-based green detection:

$$\text{mask} = \mathbf{1}\left[H \in [25, 85],\ S \in [40, 255],\ V \in [40, 255]\right] \quad (3)$$

In all cases, non-plant pixels are replaced with a neutral gray (`[240, 240, 240]`) to prevent background features from influencing classification.

[INSERT FIGURE: Segmentation Comparison]

Side-by-side: original image, FastSAM mask, SAM mask, green threshold mask.

Figure 3: Three-tier segmentation cascade results on a sample plant image.

3.2 Classification

We use `mobilenet_v2_1.0_224-plant-disease-identificat` [5], a MobileNetV2 fine-tuned on the PlantVillage dataset (∼54K images, 38 classes across 14 plant species). The model takes the segmented image, processes it through the HuggingFace `MobileNetV2ImageProcessor`, and returns a class label with confidence score.

The binary health signal is derived from the label string:

$$\text{is_healthy} = \begin{cases} \text{True} & \text{if "healthy"} \in \text{label} \\ \text{False} & \text{otherwise} \end{cases} \quad (4)$$

3.3 Grad-CAM Attention Maps

To provide visual explanations, we apply Gradient-weighted Class Activation Mapping (Grad-CAM) [6] to the last convolutional layer of MobileNetV2 (`mobilenet_v2.features[-1]`). The implementation wraps the HuggingFace model in a custom `HFModelWrapper` that extracts logits, enabling compatibility with the `pytorch-grad-cam` library.

The resulting heatmap is resized to the original image dimensions and overlaid using the standard cam-on-image visualization, highlighting regions the classifier attends to most strongly.

[INSERT FIGURE: Grad-CAM Heatmaps]

Examples showing attention maps for healthy vs. stressed plants, demonstrating focus on leaf discoloration and wilting regions.

Figure 4: Grad-CAM attention maps highlighting stress-relevant regions.

4 Stress Estimation Without Paired Data

4.1 The Core Challenge

No public dataset pairs plant photographs with simultaneously-recorded ultrasonic emission rates. Training a supervised model to predict pop rates from images is therefore not feasible with current data. This is the central methodological challenge of PlantWhisper.

4.2 Our Approach: Compositional Heuristic Pipeline

Rather than attempting to learn a direct mapping image → pops/hour, we decompose the problem into two well-studied sub-problems chained together:

$$\text{image} \xrightarrow{\text{(1) classification}} (\text{label, confidence}) \xrightarrow{\text{(2) heuristic}} \sigma \xrightarrow{\text{(3) biology}} R(\sigma) \tag{5}$$

Step 1: Classification. The MobileNetV2 classifier, trained on PlantVillage, maps the image to a disease label and confidence score. This step is fully supervised with strong performance (>95% accuracy on the PlantVillage test set).

Step 2: Stress Estimation Heuristic. We define the continuous stress level $\sigma \in [0, 1]$ as:

$$\sigma = \begin{cases} 0.1 \times (1 - c) & \text{if healthy} \\ 0.4 + 0.5 \times c & \text{if diseased} \end{cases} \tag{6}$$

where c is the classifier's confidence. The intuition:

- **Healthy + high confidence**: Very low stress ($\sigma \to 0$). The model is certain the plant is fine.
- **Healthy + low confidence**: Slightly elevated stress ($\sigma \to 0.1$). Ambiguity may indicate early stress signs.
- **Diseased + low confidence**: Moderate stress ($\sigma \approx 0.4$). Some disease indicators present.
- **Diseased + high confidence**: High stress ($\sigma \to 0.9$). Clear disease pathology.

The value is clipped to $[0, 1]$.

Step 3: Biologically-Grounded Pop Rate. We map stress to pops/hour using a **hump-shaped curve** anchored to empirical data:

$$R(\sigma) = R_{\text{healthy}} + (R_{\text{peak}} - R_{\text{healthy}}) \times 4\sigma(1 - \sigma) \tag{7}$$

where $R_{\text{healthy}} = 1.0$ and $R_{\text{peak}} = 35.0$ pops/hour. For very low stress ($\sigma < 0.1$), a linear ramp is used instead:

$$R(\sigma) = R_{\text{healthy}} + 30\sigma \quad \text{for } \sigma < 0.1 \tag{8}$$

The hump shape models the biological reality that at extreme stress ($\sigma \to 1$), the plant has already cavitated most of its vessels (hydraulic failure), so pop rate *decreases*—there are fewer functional vessels left to cavitate.

[INSERT FIGURE: Stress → Pop Rate Curve]

Plot of $R(\sigma)$ showing the hump-shaped response: peak at $\sigma = 0.5$, declining at extreme stress due to hydraulic failure.

Figure 5: Hump-shaped stress-to-pop-rate mapping, anchored to Khait et al. data.

4.3 Stress Categories

For user-facing display, we discretize stress into five tiers:

Table 2: Stress category thresholds and display colors.

Category	σ Range	Color
Healthy	$[0, 0.15)$	Green
Mild Stress	$[0.15, 0.35)$	Yellow
Moderate Stress	$[0.35, 0.55)$	Orange
Severe Stress	$[0.55, 0.75)$	Red
Critical	$[0.75, 1.0]$	Purple

4.4 Justification and Limitations

This heuristic approach is **not** a substitute for supervised learning on paired data. It is an *approximate bridge* that is:

- **Biologically plausible**: Pop rate bounds are directly from published experimental data
- **Monotonically sensible**: Sicker plants produce more pops (up to hydraulic failure)
- **Confidence-aware**: Classifier uncertainty modulates the stress estimate
- **Transparent**: The formula is explicit and interpretable, not a black box

However, it cannot capture plant-specific vulnerability curves (drought-tolerant species vs. sensitive ones), environmental context (temperature, humidity), or distinguish between types of stress that produce different pop signatures.

5 Diffusion-Based Audio Synthesis

5.1 Architecture: Conditional UNet

The core generative model is a stress-conditioned UNet for spectrogram diffusion. The architecture:

- **Input**: Noisy mel spectrogram tensor $x_t \in \mathbb{R}^{1\times 64\times 32}$
- **Conditioning**: Stress level σ and diffusion timestep t, each embedded via a 2-layer MLP ($1 \to 64 \to 64$, SiLU activation)
- **Encoder**: Three conv blocks ($1 \to 32 \to 64 \to 128$) with MaxPool2d downsampling and GroupNorm(8) normalization
- **Conditioning injection**: Additive projection of $(\sigma_{\text{emb}} + t_{\text{emb}})$ into the bottleneck feature map via a linear layer ($64 \to 128$), broadcast spatially
- **Bottleneck**: Two conv layers ($128 \to 256 \to 128$) with GroupNorm + SiLU
- **Decoder**: Three conv blocks with bilinear upsampling and skip connections (concatenation)
- **Output**: Single-channel denoised spectrogram via 1×1 convolution

5.2 Diffusion Process (DDPM)

We implement a standard Denoising Diffusion Probabilistic Model [7] with:

- $T = 500$ diffusion steps
- Linear noise schedule: $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$
- Forward process: $q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}\, x_0, (1-\bar{\alpha}_t)\mathbf{I})$

Sampling follows the standard reverse process:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x_t, t, \sigma)\right) + \sqrt{\beta_t}\,\mathbf{z} \tag{9}$$

where $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ for $t > 0$.

5.3 Spectrogram to Audio

The generated mel spectrogram is converted to audio via:

1. Exponentiate: $S_{\text{linear}} = e^{3\cdot S_{\text{generated}}}$ (undo log compression with gain)
2. Griffin-Lim inversion via `librosa.feature.inverse.mel_to_audio` with STFT parameters: `n_fft=256`, `hop_length=64`, 32 Griffin-Lim iterations at $f_s = 22{,}050\,\text{Hz}$

The resulting single-pop audio is then placed into a full-duration clip at randomized positions matching the predicted pop rate.

5.4 Synthetic Fallback

When the diffusion model checkpoint is unavailable, a synthetic fallback generates pops using:

$$\text{pop}(t) = \mathcal{N}(0, 0.5) \cdot W_{\text{Hanning}}(n_{\text{pop}}) \tag{10}$$

where each pop is 2 ms of Hanning-windowed Gaussian noise, placed at random positions in a 10-second clip with low-level background noise ($\sim$0.01 amplitude).

6 Waveform and Spectrogram Visualization

To provide users with visual evidence of the synthesized acoustic emissions, we generate two server-side rendered plots using Matplotlib (Agg backend for headless operation):

6.1 Waveform Display

The raw audio waveform is plotted with time on the x-axis and amplitude on the y-axis. Pop events are automatically detected as amplitude spikes exceeding 4× the signal's standard deviation:

$$\text{pop_mask}[n] = \mathbf{1}\left[|x[n]| > 4 \cdot \text{std}(x)\right] \tag{11}$$

Detected pops are highlighted with red scatter points overlaid on the green waveform trace. The plot title includes the predicted pop rate for quantitative context.

6.2 Spectrogram Display

A time-frequency representation is computed via STFT with parameters matched to the audio generation (`NFFT=256`, overlap of 200 samples), displayed using the "magma" colormap on a dB scale. The frequency axis is limited to 0–3 kHz to focus on the region containing the frequency-shifted pop energy.

Both plots use a dark color scheme (`#0a1a0f` background, emerald axis labels) consistent with the frontend design, and are transmitted as base64-encoded JPEG images.

[INSERT FIGURE: Waveform & Spectrogram]

Example output showing waveform with detected pops (red) and spectrogram with visible pop energy bursts at ∼1 kHz.

Figure 6: Generated waveform (top) and spectrogram (bottom) for a moderately stressed plant ($\sigma = 0.47$, 33.6 pops/hour).

7 Plant Voice: LLM + TTS Pipeline

7.1 Empathetic Speech Generation

PlantWhisper gives each plant a "voice" using a two-stage pipeline:

Stage 1 — LLM Text Generation. We use Llama 3.3 70B (via Groq API, model `llama-3.3-70b-versatile`) with a carefully crafted system prompt that instructs the model to speak as a conscious plant:

- First-person perspective
- Mood calibrated to stress level ("content" at $\sigma < 0.15$, "desperate" at $\sigma > 0.75$)
- Natural biological references (water transport, xylem, leaves, roots)
- Mention of ultrasonic clicks when stressed
- 2–3 sentences, no emojis, temperature 0.85

When the Groq API is unavailable, three hard-coded fallback messages cover the stress spectrum.

Stage 2 — Text-to-Speech. Microsoft Edge TTS (`en-US-AriaNeural`) renders the text with prosody matched to stress:

Table 3: TTS prosody parameters by stress level.

Stress Range	Rate	Pitch
$\sigma < 0.3$ (calm)	−5%	−2 Hz
$0.3 \leq \sigma < 0.6$ (moderate)	+0%	+0 Hz
$\sigma \geq 0.6$ (distressed)	+10%	+3 Hz

7.2 Care Recommendations

A second LLM call (or fallback template) generates 3–5 actionable care recommendations based on the stress category. This provides practical value beyond the artistic sonification.

8 Failure Cases and How We Handled Them

8.1 Segmentation Failures

Problem. FastSAM sometimes returns no masks for small plants against complex backgrounds, or segments background objects instead of the plant.

Solution. The three-tier cascade (FastSAM → SAM → green threshold) ensures graceful degradation. The center-bias heuristic for mask selection (Eq. 2) handles multi-object scenes by assuming the subject is approximately centered.

Remaining limitation. Non-green plants (red coleus, brown dormant plants) fail the green threshold fallback entirely.

8.2 Classification Domain Mismatch

Problem. MobileNetV2 was trained on PlantVillage (lab-curated images of 14 crop species). Real-world photos differ dramatically: varied lighting, multiple plants, soil visible, pots, backgrounds, non-crop species.

Solution. Segmentation removes background noise before classification. However, species outside the training set (houseplants, trees, succulents) receive unreliable labels, though the binary healthy/diseased signal often remains directionally correct.

Remaining limitation. Confidence calibration is poor out-of-distribution. The model may be highly confident about an incorrect label, propagating error through the stress estimate.

8.3 Stress Estimation Without Ground Truth

Problem. No paired image–pop-rate data exists. Our heuristic (Eq. 6) is an educated guess, not a learned mapping.

Solution. We designed the heuristic to be:

- Bounded by empirical data (Khait et al.)
- Monotonically consistent with biological expectation
- Confidence-modulated (uncertain classifications yield moderate, not extreme, stress)
- Interpretable and auditable

Remaining limitation. The system cannot distinguish between drought stress (which causes cavitation) and nutrient deficiency or pest damage (which may not), yet maps all disease indicators to the same acoustic output.

8.4 Diffusion Model Quality

Problem. Training the DDPM on actual ultrasonic recordings would require a large dataset of plant-emitted sounds. No such public dataset exists at sufficient scale.

Solution. The diffusion model is trained on synthetic spectrograms parameterized by stress level, making it a *generative prior for the acoustic model* rather than a learned representation of real plant sounds. When the checkpoint is unavailable, the synthetic fallback (Hanning-windowed noise bursts) provides a serviceable approximation.

Remaining limitation. Generated audio is perceptually plausible but not physically accurate. Real cavitation pops have complex spectral characteristics shaped by vessel geometry that our model does not capture.

8.5 TTS and LLM Availability

Problem. Both Edge TTS and Groq API require network access and may be rate-limited or unavailable.

Solution. Both have complete offline fallbacks: hardcoded speech text (3 stress-level variants) for the LLM, and the API endpoint includes async TTS fallback that retries at the API layer if the backend's synchronous event loop fails.

8.6 Memory and Latency

Problem. Loading MobileNetV2 + FastSAM + SAM + Diffusion UNet simultaneously on CPU exceeds the 16 GB RAM limit of free HuggingFace Spaces.

Solution. Lazy loading with availability flags. Each model load is wrapped in `try/except`, setting `FASTSAM_AVAILABLE`, `SAM_AVAILABLE`, `DIFFUSION_AVAILABLE`, etc. The system runs with whatever models fit in memory, degrading gracefully.

[INSERT FIGURE: Failure Case Examples]

(a) Segmentation failure on non-green plant. (b) Misclassification of out-of-distribution species. (c) Green threshold fallback result.

Figure 7: Representative failure cases and fallback behavior.

9 Deployment Architecture

9.1 Backend

The backend is containerized using Docker (Python 3.11-slim base) with:

- **Runtime**: Uvicorn ASGI server on port 7860
- **Framework**: FastAPI with full CORS support
- **Image format**: JPEG encoding at quality 85 for bandwidth efficiency
- **Audio format**: WAV (soundfile) → base64 data URIs
- **Visualization**: Matplotlib Agg backend, JPEG output
- **System dependencies**: libgl1, libglib2.0, libsndfile1, ffmpeg

9.2 Frontend

Next.js 14 with:

- Single-page React application (`"use client"`)
- Glassmorphic UI with emerald color palette
- Animated floating gradient background
- Responsive layout with CSS grid
- All data received as base64 data URIs (no file downloads)

9.3 API Contract

A single `POST /api/analyze` endpoint accepts a multipart image upload and returns a JSON response containing all results:

Table 4: API response fields.

Field	Type
`stress`	float $\in [0, 1]$
`category`	string (5 tiers)
`popsPerHour`	float
`label`	string (disease label)
`confidence`	float
`speech`	string (plant voice)
`recommendations`	string (care advice)
`voiceAudio`	data URI (MP3)
`ultrasonicAudio`	data URI (WAV)
`diffusionAudio`	data URI (WAV)
`segmentedImage`	data URI (JPEG)
`gradcamImage`	data URI (JPEG)
`waveformImage`	data URI (JPEG)
`spectrogramImage`	data URI (JPEG)

10 Results and Discussion

10.1 Qualitative Observations

PlantWhisper produces compelling multi-modal outputs:

- Healthy plants yield quiet waveforms with sparse, low-amplitude pops (∼1/hour)
- Diseased plants produce dense, energetic pop patterns clearly visible in both waveform and spectrogram
- Grad-CAM consistently highlights discolored, wilting, or spotted leaf regions
- LLM-generated speech meaningfully reflects stress level (calm vs. distressed tone)

10.2 What This System Is and Is Not

PlantWhisper is:

- A **creative exploration** of how AI can bridge modalities (vision → sound)
- An **educational tool** demonstrating real plant biology (cavitation acoustics)
- A **proof of concept** for multi-modal plant monitoring
- A **portfolio piece** showcasing full-stack ML engineering

PlantWhisper is **not**:

- A validated agricultural diagnostic tool
- A replacement for actual ultrasonic sensors
- Trained on real plant acoustic data
- Suitable for making agricultural decisions without domain expert review

11 Future Work

1. **Paired dataset collection**: Deploying contact microphones alongside cameras on greenhouse plants to build the first image-to-pop-rate dataset
2. **Species-specific models**: Vulnerability curves differ by species; a species classifier could select appropriate stress-to-pop mappings
3. **Temporal analysis**: Video input enabling time-series stress tracking
4. **Real ultrasonic training data**: Training the diffusion model on actual recorded cavitation events
5. **Multi-sensor fusion**: Combining thermal, hyperspectral, and acoustic sensors with visual data

[INSERT FIGURE: Full Pipeline Example — Two-Column Width]

Complete output for a healthy and a stressed plant side-by-side: original image, segmented image, Grad-CAM, waveform, spectrogram, stress gauge, speech text.

Figure 8: Complete PlantWhisper output for a healthy plant (left) vs. a severely stressed plant (right).

12 Conclusion

PlantWhisper demonstrates that meaningful cross-modal inference—from a photograph to a sound—is possible when grounded in real biology and transparent heuristic design. By combining established computer vision models with biologically-anchored acoustic modeling and a conditional diffusion model, we create an end-to-end system that makes the silent stress of plants audible. The system is honest about its limitations: without paired training data, the acoustic output is an approximation, not a measurement. But as both an educational tool and a technical proof of concept, PlantWhisper opens a window into a world of plant communication that humans cannot normally perceive.

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A Reproducibility

All source code is available at https://github.com/Iammohithhh/plantwhisper. The backend runs on any machine with Python 3.11+ and the listed dependencies. No GPU is required. The diffusion model checkpoint (`diffusion_model.pt`) can be trained using the provided architecture or omitted to use the synthetic fallback.